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Leveraging Smart Meter Technology to Combat Energy Theft and Empower Consumption Monitoring

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***Abstract*— Monitoring electricity theft is an integral part in reducing energy losses in electrical power systems. Electricity losses can arise at all levels, starting at generation, followed through transmission and distribution. The distribution level is generally where higher percentage of avertable losses are observed. Mostly, all the electrical energy distribution suppliers operate with losses that are accepted within a certain range. In electrical power systems, losses fall into two types, and they are determined by the components that are either technical or non-technical. Technical losses occur due to physical characteristics of the power system such as resistance, leakage and corona discharge and Non-technical losses occurs due to human action or external factors such as faulty meters, electricity theft and billing system efficiency. In the smart grid, AMI's real-time data stream empowers utilities to fine-tune operations, predict problems, and effortlessly integrate renewable sources, resulting in a smarter, more robust, and efficient electricity system for everyone. It is, however, a difficult task as well to put together AMI secure due to its complex network because of its vast, diverse network such as multiple entry point cyberattacks. Technical losses occur mainly due to power supply dissipation which ranges from 6% to 7%. These may be brought on by aging-related defects in the distribution and transmission lines, malfunctioning transformers and measuring systems, corona discharge, and insulator failure. These are the results of internal actions of the power system network. Unauthorized consumption, such as meter tampering and bypassing connections, leads to substantial non-technical energy losses, ranging from 15% to 30% across different states. It is practicable to monitor the energy thefts in the distribution network by means of developing a smart meter such as Tamper Detection, Data Collection, Unusual Pattern Identification.**

Keywords*— *Energy theft, Smart grid, smart meter, Radio Frequency Identification, Microcontroller, Internet of Things, Data security, Energy losses, Distribution supply.

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I. INTRODUCTION

In India, in the present electricity meters, which are basically electromechanical in nature, restrict the electrical energy suppliers to track and record the varying maximum demand of consumers. The consumers experience problems regarding their consumption. Monitoring the load of consumers on a fixed periodic basis is the best way to address all of these issues. This might also be used to measure maximum demand, provide accurate bills, and find instances of power theft. The mentioned are the features to be taken into consideration in order to design an efficient energy metering system. The paper explores the creation of an integrated energy meter as a potential solution to the problems that the current system is causing for distribution corporations and customers alike. The paper mainly deals with smart energy meter, which is used to implement the desired functionality by utilizing the features of embedded systems. It focuses on creating a smart meter that can identify power thefts occurring within the network. In the earlier phase of technology, traditional meters have been used in terms of electricity distribution. Traditional meters are employed to measure the electricity consumption in every household. The traditional meters can be manipulated in various ways to consume electricity unauthorized. Also the users do not get the complete access to view their usage and contact the electricity supplier. As a result, the use of traditional electricity meters has been declining lately. Moreover, traditional electricity meters require the assistance of human workforce to manually check for the usage and generate bills. To overcome these, traditional meters can be upgraded to smart meters [2]. These collects data of the consumption and the data is shared to the supplier as well the consumer to analyse the usage and irregularities like energy thefts. In the present world, electromechanical meters are replaced by digital meters. These could be a more convenient method for reading energy usage or consumption.

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For the wireless transmission of data, various wireless technologies could be used like Bluetooth, ZigBee, GSM and so on. The two potential non-human data collection techniques are used. Communication networks are used for the transfer of data back and forth in the smart grid technology. Smart meter technology enables the collection of consumption data on a timely basis. This data collected is used to study the nature of consumption by the consumers and thus helps in detection of any unauthorized usage or manipulations. Moreover, there is less human intervention as most of the functions are automated. Therefore, the investment cost is reduced with improved efficiency and security. This can reduce the annual expenditure of the utility centers to a great extend [3].

II. Literature Review

Electricity theft is found to be a very general issue in many of the developing countries usually where there is an extremely high demand for electricity [4]. According to researches and surveys, the Sri Lankan power station itself experiences loss of around 30% of its overall electricity distribution system. Electricity losses can be caused by various reasons. Transmission losses can vary from 2% to 8% based on their voltage level and distance. The major two reasons include electricity theft and transmission losses. The transmission losses can be controlled by using more efficient Equipments for transmission and distribution. It is recognized that electricity theft is often achieved by performing actions such as illegal connections, tampering of the meter, unpaid bills and billing discretion, through the studies conducted by Smith [5]. A research conducted between 1989 and 2010 across over 102 nations found that the incidence of electricity theft is rising dramatically. From the studies, it is observed that electricity theft take place at two places which are residential energy meter or electricity meter and distribution lines. When it comes to household energy meter, the electricity theft is carried out by means of sidestepping. Sidestepping of the energy meter is performed by using simply a piece of wire and using it to bypass the energy meter.

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To determine which theft occurred in the distribution line, the fed in electricity and output electricity consumption of the selected poles are compared. Any deviation of the value observed in the comparison procedure implies that the pole has drawn excess current which ultimately implies the occurrence of electricity theft. To achieve this, two transformers are placed in such a way that the values of electricity can be observed across two ends. Electricity transformers are placed at two sides, one at the electricity line and another one at the end of the residential lines. Fault in both transmission and distribution lines can contribute to technical losses by causing temporary disruptions and increasing resistance in the power lines, leading to energy dissipation in the form of heat [18]. Minimizing these losses is essential for ensuring the reliable and efficient delivery of electrical power to end-users [16] [17].

The values are processed by the microcontroller which converts the analog inputs into digital signals using analog to digital converters. [18] The current and voltage are measured periodically and the data is stored in a served. On comparison of the current and voltage values across the pole and the consumer end, if found any deviation, then it is treated as energy theft. When the electricity theft is detected by the system, it makes use of GSM module to send the information to the supplier and in turn the board can take the necessary action to prevent power loss. Furthermore, the proposed system is can send the respective readings of electricity meter to the power supplying board, automatically which in turn eliminates the need for the authority individual to manually visit the consumer sight to take readings. Thus, the labor costs are reduced.

III. Research Methodology

In the existing method, the unauthorized use of electricity, commonly known as electricity theft, has negative financial consequences for utility companies. The electricity meters aren't capable of detecting electricity theft. Some of the limitations when it comes to using the traditional electromechanical meters are that consumers do not have access to view their consumption. Periodic checks are not always completely efficient and effective due to its arduous nature [6]. Adopting to a smart meter can be effective as it’s an integrated device that can simultaneously measure the consumption and detect electricity thefts without any human intervention. It is a more precise and efficient way to read power meters in isolated locations that might not be reachable. The current system's drawbacks include lower security and less precise meter readings in remote locations. Requires human intervention which ultimately increases the expenditure.

The recommended system includes real-time power monitoring and control, along with the capability to detect instances of electricity theft. The recommended system is proceeded with the development of a smart electricity meter capable of real time electricity monitoring and theft detection in the distribution networks [11]. The energy theft is minimized in the distribution network by including technological measures such as smart meters, distribution network monitory, pre-paid metering systems and non-technical measures such as penalties. The block diagram shown in Fig 1 that illustrates how the suggested system's tasks would flow.

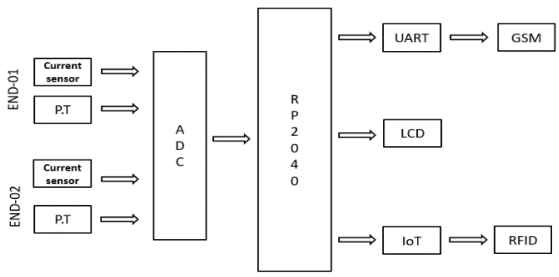

Fig.1. Block diagram of Energy Theft Detection

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This proposed work consists of a current sensor and a potential transformer at either ends of the electricity meter, which are used to measure current and voltage, respectively. Consider End 1 as the side before the meter and End 2 as the one after the meter [7]. At End 1, the voltage and current are measured, respectively, by voltage transformers and current sensors. Similarly, End 2 measures the current and voltage that flows beyond the electricity meter. The complete control of the system lies within the RP2040. The RP2040 is a microcontroller integrated circuit with 32-bit dual ARM Cortex-M0+. It was developed by Raspberry Pi Ltd, officially in the year 2021[8][1]. It is low cost and support various programming languages such as Assembly, C/C++ and Python. It delivers high performance with less power consumption [9].

The current and voltage are measured continuously, and is updated to the server database. This in turn is developed for detecting electricity theft and the real time monitoring of the energy received and consumed by the user. The obtained values from the two ends between the consumer and the pole are checked for difference. If no difference is present, then it implies that the current flow is theft free. If an external circuit draws current from the electricity line, which is an act of illegal consumption of electricity, it is considered as energy theft. On occurrence of an energy theft, the current and voltage values of the two ends do not match. The system is programmed to notify the respective utility supplier immediately on occurrence of the theft along with the location. This is carried out successfully by the UART followed by the GSM module and GPS. The GPS is fixed at every node where the branching of the electricity lines occurs. The location of theft could also be identified using the meter id. Every smart meter has been registered with a unique address with respective location. Hence, the respective address and location of the node is updated in the application so that the electricity supplier can be aware of theft and can take necessary actions to prevent losses [10].

The proposed system is inclusive of an RFID based payment portal where the billing details are automatically performed and the user can directly pay from their own premises. Beyond barcodes, RFID technology offers a versatile solution for diverse business needs. RFID empowers businesses to streamline operations, reduce costs, and improve asset utilization by automatically identifying and tracking assets, leading to enhanced efficiency and quality across various industries [12]. RFID forms the heart of Internet of things. It could be used to manifest efficient asset management system. The advantages of RFID include: RFID automates collection of data and therefore human effort and errors are reduced to a higher extend. It can support the reading of tag with no line-of-sight, Numerous RFID tags can be read simultaneously, therefore provides an increased efficiency, mitigate human errors RFID and also eliminates the likelihood of human error. The next most recognizable and immensely beneficial feature of the system is the automated relay ON/OFF system. The RP2040 is programmed to automatically turn off the supply to the user when he/she fails to pay their electricity bill [13]. This feature turns out to be one of the most advantageous one when it comes to economic terms. The electricity supplier can completely negotiate the investment in employees who are involved in performing this task. Thus, the labor charges are eliminated, benefitting the supplier [14] shown in Fig 2.

The use of laser or infrared sensor is employed to prevent the manual exploitation of the smart electricity meter [15]. At times when the unauthorized individuals try to manipulate the system, the infrared sensor detects it and notifies the electricity authority.

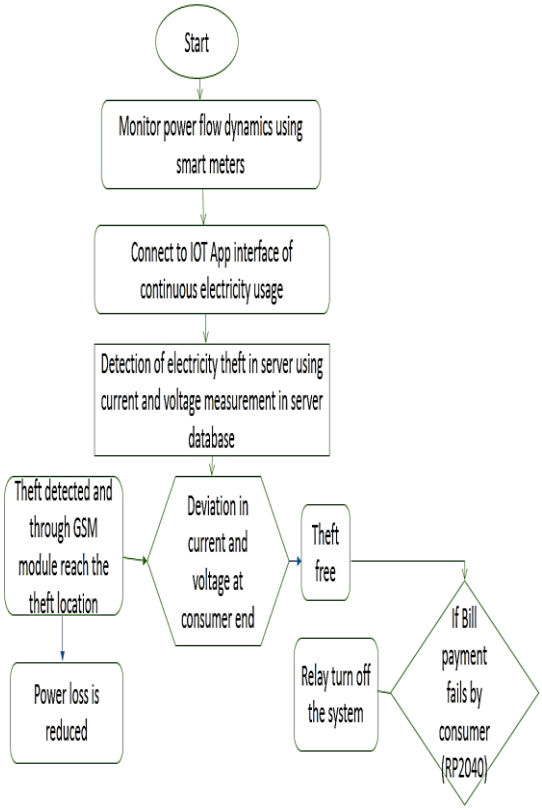

Fig.2. Workflow of Energy Theft Detection

IV. Results And Discussions

This proposed system integrates the model to detect energy theft and monitor the power usage. The main features include:

1. Continuous monitoring of electricity usage.
2. Energy theft detection along with the location.
3. Bill payment using RFID.
4. Automatic relay off when bill not paid.

The simulation of the proposed system is carried out using the Proteus 8.11v. The main purpose of the Proteus Design Suite software is to automate electrical design. It is a great tool for the electronic design engineers and technicians. It is widely used in industrial sectors to generate schematics for circuit board manufacturing and PCB design. It serves as a platform that could be used for the complete design and implementation of circuit boards. Its advantages include intelligent layout, hybrid circuit simulation and accurate analysis, efficient software debugging, PCB automatic layout. The proposed system is simulated using the software Proteus 8.11V. The LCD

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displays the current and voltage and the supply end and the receiver end, continuously.

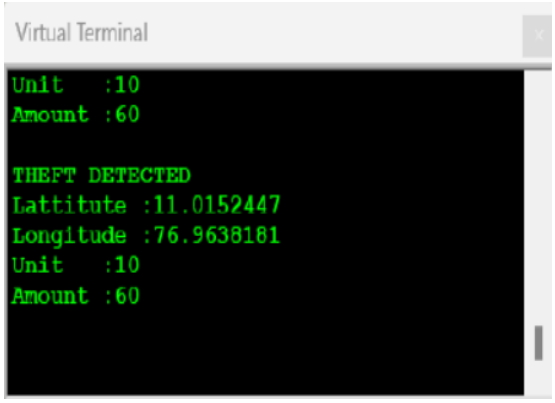

Fig.3. After the Electricity theft detected along with the location

The simulation results for the electricity theft detection is shown in Fig 3. The external unauthorized circuit becomes active. Immediately, the theft is detected and displayed along with the location. Once the external circuit becomes inactive, the circuit goes back to normal and the supply flows uninterrupted.

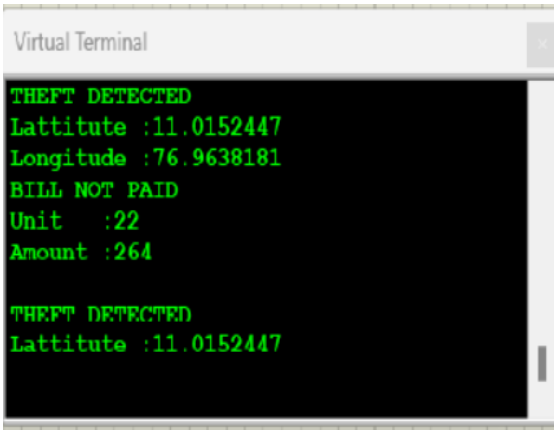

Fig.4. Bill Payment- Automatic relay OFF

The Fig 4 represents the automatic relay off when the user fails to pay the electricity bill. This eliminates the need for extra labour and cost. The finished hardware working model of the proposed system has been developed which is inclusive of the features stated. The Fig 5 illustrates the end product of the suggested system.

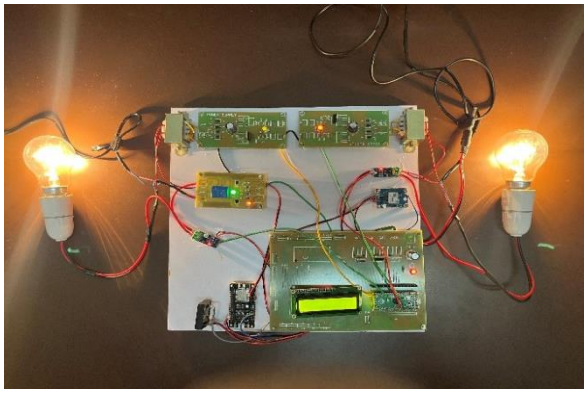

Fig.5. Hardware implementation

The results have been obtained for the same. The Fig 6 displays the current and voltage measurements; the Fig 7 shows the bill amount paid.

Fig.6 Current and voltage monitoring continuously

Fig.7. Displays the electricity bill amount paid

An Blynk application is developed for the same which provides a user friendly interface to interact with the system which is easy to use is shown in the Fig 8.

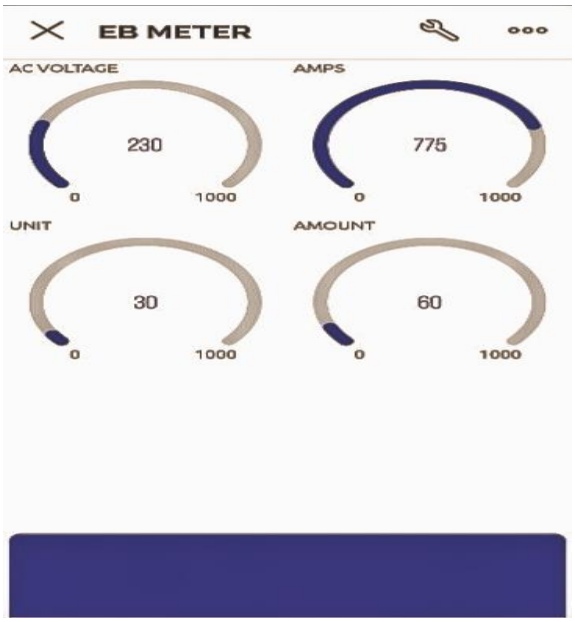

Fig.8. IoT App interface of continuous electricity usage

The created application's interface, which indicates the constant power use, is seen in Fig 8. The app is developed

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to display the current and voltage and the supply end and the receiver end, continuously.

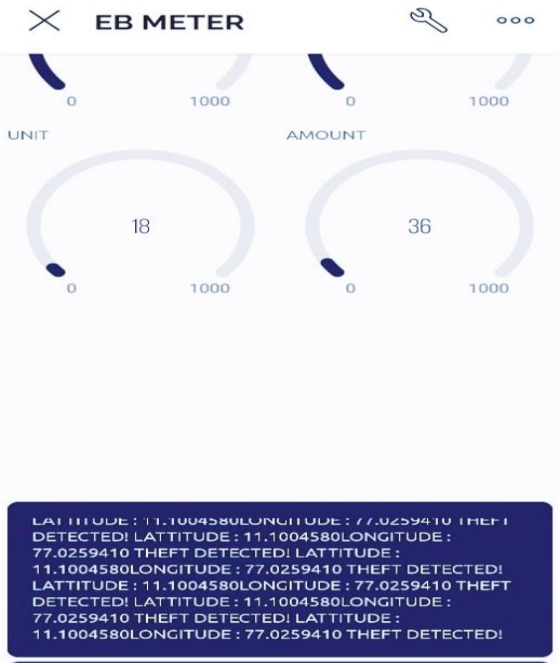

Fig.9. Electricity theft detected along with the location

The results obtained for the detection of electricity theft is shown in the Fig 9 . The external unauthorized circuit becomes active. Immediately, the theft is detected and displayed along with the location in the application developed. Once the external circuit becomes inactive, the circuit goes back to normal and the supply flows uninterrupted.

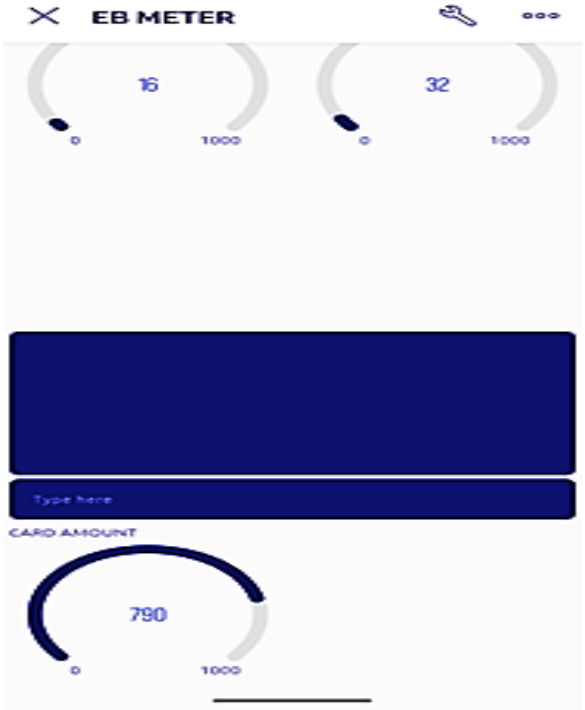

Fig.10. RFID based bill payment

The Fig 10. represents the RFID based bill payment. The proposed system also comes with a feature that automatically turns the relay off when the user fails to pay the electricity bill. This eliminates the need for extra labour and cost.

V. Conclusion

In developing countries like India, traditional meters could be replaced using sophisticated and more accurate smart meters that automates the almost the entire smart grid network. A high amount of electricity is lost as a result of electricity theft, incorrect meter reading and billing.The electricity losses are almost unfeasible to be measured and analyzed by means of traditional power system analysis tools. The smart meter system makes sure that the measurements are accurate and supports the reliable estimation of power consumed. Moreover, it almost eliminates the manual works. It is an integrated system that combines real time monitoring and control of electricity with electricity theft detection in the most feasible and effective way. The future work can explore how 5G and IoT can enhance smart meters by enabling features like dynamic pricing, remote diagnostics, and proactive maintenance.

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